

Technology Stack

AI-Assisted Smart Budget Planner

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Responsive static interface without a frontend framework.
2	Backend / Server-Side	Python 3.11+, FastAPI, Uvicorn	API application and web serving.
3	Data Validation	Pydantic v2	Typed request/response models.
4	Database / Storage	SQLite	Stores users and saved plans.
5	Authentication	PyJWT + HTTP-only cookies	Bearer JWT support and browser sessions.
6	Password Security	scrypt	Salted password hashing.
7	AI Integration	Google Gemini gemini-2.5-flash + HTTPX	Optional one-line style/tradeoff commentary.
8	Testing	Pytest	Automated application tests.

The project report describes Gemini as optional and states that the deterministic planner, not the AI model, controls the arithmetic and budget-cap guarantee.